

SatQuery AI - Multimodal Remote Sensing Evidence Report

Autonomous Agentic Execution & Evidence Provenance Trace

Query:	"What type of land cover dominates the scene?"
Task Category:	VQA_CAPTION
Specialist Tool:	RS-VQA & Scene Captioning Tool (BigEarthNet Adapted)
Reliability Score:	60.5% (LOW TRUST)
Execution Time:	58.2 ms

1. Agent Synthesis & Visual-Language Answer

Remote Sensing VQA Analysis Result:

- Primary Visual Land Cover: Inland wetlands (Marshes, Peat bogs) (model probability: 25.0%).
- Associated Features: Industrial, commercial and transport units (15.1%) and Mine, dump and construction sites (5.8%).
- Multimodal Grounding: Answer generated by cross-conditioning image visual embeddings with text query feature representations.

2. Auditable Agent Pipeline Execution Trace

- **[Perception Layer]** Format & CRS validated. Mode: Single Optical Image (1 image(s)). Status: PASSED
- **[Query Engine]** Classified intent: 'VQA_CAPTION'. Extracted entities: ['general_land_cover']. Task: 'vqa_caption'
- **[Mission Planner]** Selected tool: 'RS-VQA & Scene Description Specialist (BigEarthNet Adapted)' with params {'top_k_classes': 5}. Generated 7 node task-graph.
- **[Specialist Tool Execution]** Executed specialist tool 'RS-VQA & Scene Captioning Tool (BigEarthNet Adapted)' with model confidence 45.0%.
- **[Evidence Fusion]** Aligned 2 evidence streams. Deep visual feature distribution aligns with BigEarthNet land-cover taxonomy.
- **[Trust & Uncertainty]** Calculated Reliability Score: 60.5% (LOW TRUST). Cross-model agreement: 50.0%.
- **[Discovery Engine]** Autonomous scan completed: Autonomous scan identified 1 'Beyond Your Query' discovery item(s).
- **[Answer Generator]** Final evidence-linked response synthesized in 58.2 ms.